Yitao Zeng

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Education

Southern University of Science and Technology (SUSTech)- Automation
2023–Present

- **GPA:** 3.87/4.0（**1/35**）

Relevant Coursework: Advanced Control for Robotics (Grad), Robotic Motion & Control, AI & Machine Learning, Optimal Control & Estimation, Signals & Systems, Data Structures & Algorithms.

Chengdu No.7 High School
2020–2023

Skills

- **Embodied AI:** Proficient in VLA, Reinforcement Learning, and Machine Learning with practical application experience.Solid understanding of robotic kinematics and dynamics; experienced with robotic deployment pipelines both in sim and real.
- **Frameworks & Tools:** Skilled in ROS, wandb; experienced with simulation tools including Isaac Gym/Lab/Sim, MuJoCo
- **Programming Languages:** Python, Pytorch (extensive research experience), C++/C/Java/Matlab/Arduino (course project)

Research

Vision Language Action Models with Synthetic Data

Research Intern, Embodied Perception and InteraCtion (EPIC) Lab / Galbot, Peking University (2026.1 – Present)

- Conducted baseline evaluation of 3D Diffusion Policy trained exclusively on synthetic data.
- Deployed the policy on a Franka robot arm with ZED Mini stereo camera for real-world validation.
- Accelerated the migration of a 20TB dataset from TFDS to LeRobot format by **leveraging high-performance computing (HPC) for parallelized data transformation**."

Robotic Arm Manipulation via 3D Diffusion Policy

Research Assistant, Control and Learning for Robotics and Autonomy (CLEAR) Laboratory, SUSTech(2025.2-2025.12)

- Led an end-to-end pipeline for robotic manipulation, with key responsibilities including:
  - Data collection, camera calibration, image preprocessing.
  - Training and deploying the robot on both Franka arm and ARX (方舟无限).

Pointfoot and wheelfoot robot locomotion with reinforcement learning

Research Assistant, Control and Learning for Robotics and Autonomy (CLEAR) Laboratory, SUSTech(2025.10-2026.1)

- Developed and tuned locomotion controllers for point-foot and wheel-foot robots using deep reinforcement learning (DRL) in the Isaac Gym simulator.
- Achieved successful sim-to-sim and sim-to-real deployment, enabling a wheel-foot robot to stably climb 18cm stairs and traverse diverse terrains in mujoco and in the real world.

Roboverse Contributions (Remote)

Research Assistant, Berkeley Artificial Intelligence Research (BAIR) Laboratory, University of California, Berkeley (2025.7-2025.9)

- Maintained the RoboVerse codebase and successfully contributed via pull requests.
- Migrated and validated SkillBlender tasks within RoboVerse to ensure environmental and benchmarking parity.

Awards & Honors

- **First Prize Scholarship(rank 3/188)**, SUSTech
- **Da Vinci Challenge Camp**, *Competition, SUSTech (July 2024)*
    - **Platinum Award** (Rank 1 out of 6 teams) & **Sole "Best Functional Design Award"**
    - Designed and built an autonomous vehicle actuated through Arduino language with a robotic claw for block retrieval; independently led the navigation and control module.
- **Tron Camp**, *Competition, LimX Dynamics (July-August 2025)*
    - **Sole "Outstanding Team Award" recipient (the youngest finalist)**
    - Independently devised a reinforcement-learning control strategy enabling a wheel-legged robot to climb stairs in a real environment.
- **Chinese National Challenge Cup**, *Competition, Ministry of Education of Guangdong (July-August 2025)*
    - **Provincial Gold Medal (rank 4th in the name list)**
    - Played a pivotal role in the project's final phase by refining the technical narrative and presentation of a **robotic fishing vessel** project,, which was instrumental in securing the provincial gold medal.
- **National Foreign Language Contest in Translation** (English):
    - **First place**, SUSTech (2023) **Silver Award**, Guangdong Provincial Round (2024)
- **International Communication Short Video Contest** (English):
    - **2nd Place**, SUSTech (2024) **Bronze award**, Guangdong Provincial Round (2025)

Hobbies and student organizations

Sports

- Swimming： Earned **First-Class Athlete (School Level)** in **Freestyle & Butterfly** (**2nd Place** in both events at the school competition)
- Soccer Freshman Cup: Vice-captain, 2nd Place
- Soccer Inter-College Cup: 2024: 3rd Place， 2025: 2nd Place

Piano

- Performance of "He's a Pirate" at an ACG (animation, comics, and games) theme event (2023).
- Performance of "Flower Dance" at the Piano Recital for the Culture and Arts Week in SUSTech(2024).

Student Red Cross Society, SUSTech

- **Member, Organization Department** (2023 - 2024)
- **Member, First-Aid Department** (2024 - 2025)
- Accumulated over **230 volunteer service hours** (2023 - 2025)